



COMPONENTS

Battery management IC

Infineon Technologies has expanded its battery management systems (BMS) portfolio with new sensing and balancing IC, TLE9012AQU. The device



is mainly designed for batteries in hybrid and electric cars. It

measures the voltage in up to twelve battery cells with an accuracy of $\pm 5.8\text{mV}$ over the entire temperature and voltage range as well as the operating life cycle.

Furthermore, it supports up to five external temperature sensors and uses an iso-UART interface for communication.

The TLE9012AQU provides the necessary measurement data and ensures a balanced state of charge through cell balancing. Cell balancing is achieved via twelve balancing switches integrated on the chip—one per channel. These are designed for currents up to 150mA. For higher balancing currents, the device also supports external switches.

With its twelve channels, the TLE9012AQU is particularly suitable for batteries that are partitioned with twelve cells per module. One device is then required per module.

Infineon Technologies
www.infineon.com

BLE module

STMicroelectronics has launched the STEVAL-IDB008V1M Bluetooth Low Energy 5.0 (BLE) evaluation platform for accelerating application development with modules featuring BlueNRG-2, the company's second-generation BLE System-on-Chip (SoC).



BlueNRG-2 supports Bluetooth 5.0 certification, which allows enhanced security with LE Secure Connections,

power-efficient privacy with Link Layer Privacy 1.2, and up to 2.6-times higher throughput with LE Data Length Extension.

The SoC contains an Arm Cortex-M0 core operating at up to 32MHz to handle Bluetooth stack and application processing. It features 32kHz ring oscillator, 24kB RAM, and 25kB flash program memory.

The BlueNRG-M2SA module combines a compact 13.5mm $\times$ 11.5mm form factor BlueNRG-2 SoC with an efficient ceramic antenna, RF balun circuit, 32kHz crystal oscillator, and SMPS inductor to further reduce power consumption.

The STEVAL-IDB008V1M plug-and-play kit enables rapid evaluation and product development using the BlueNRG-M2SA module.

STMicroelectronics
www.st.com

Data security IC

NXP Semiconductors has launched the new MIFARE DESFire EV3 IC that brings in next-generation performance,



advanced security, and seamless integration of mobile services for providing better security and connectivity in smart city services.

As the third upgraded version of NXP's contactless MIFARE DESFire portfolio, the latest IC is backward compatible and offers enhanced performance with greater operating distance and improved transaction speed.

In combination with its advanced security features, the new IC delivers faster and more secure contactless transactions such as paying for parking, accessing offices or campuses, and using other essential city services.

The new transaction timer feature helps mitigate a man-in-the-middle attack, making it harder for an attacker to interfere with the transaction.

NXP Semiconductors
www.nxp.com

Film capacitors

KEMET has launched three new series of metallised polypropylene dielectric film capacitors: C44U-M, C44P-R, and R75H. The latest C44U-M series is designed for high-power conversion systems in green energy, industrial, and energy storage systems. It supports DC networks and



supplies high current for circuit needs or load demands in wind, solar, and storage

energy technologies. High DC voltage load capability (1,800V DC) and high ripple current make this UL810-certified capacitor optimal for high capacitance while minimising equivalent series resistance (ESR).

Similarly, the new C44P-R series is also UL810-certified and features high current capability for input and output AC filtering (up to 1,000V AC) in solar converters, wind turbines, uninterruptible power supplies, and power factor correction systems. These cylindrical aluminium canister capacitors allow higher energy density, extended life, and reliability.

Designed with a radial box style for PCB mounting, the R75H Pulse series includes the same high energy density, extended life, and reliability benefits. This series meets one of the most rigorous reliability standards in the industry with its high voltage range (up to 2,000V DC), high capacitance range, and high dv/dt capability.

KEMET
www.kemet.com

UV LEDs

The KLSL3535UC UVC LEDs from LEDchip Indus is ideal for compact and